

Sample Paper C:

Q2: Hydrogen atom: $\psi(r, \theta, \phi) = C r e^{-kr} \sin(\theta) e^{-i\phi}$

(a) Here we are simply applying the operators $\hat{L}^2$ and $\hat{L}_z$ which they give you on pg 3

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$
$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \right]$$

→ For ψ to be an eigen state then we need

$$\hat{L}_z \psi = \hbar m \psi, \quad \lambda_1, \lambda_2: \text{Values we must find}$$
$$\hat{L}^2 \psi = \hbar^2 l(l+1) \psi$$

only acts on ϕ

$$\hat{L}_z: \hat{L}_z \psi = -i\hbar \frac{\partial}{\partial \phi} (C r e^{-kr} \sin(\theta) e^{-i\phi})$$
$$= -i\hbar C r e^{-kr} \sin(\theta) \frac{\partial}{\partial \phi} (e^{-i\phi})$$
$$= -i\hbar C r e^{-kr} \sin(\theta) (-i) e^{-i\phi}$$
$$= (-\hbar) (C r e^{-kr} \sin(\theta) e^{-i\phi})$$

ψ

$$\hat{L}_z \psi = (-1)\hbar \psi$$

$$\Rightarrow m = -1$$

$$\hat{L}^2: \hat{L}^2 \psi = -\hbar^2 \left[\underbrace{\frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \psi}{\partial \theta} \right)}_{\substack{\text{this term only} \\ \text{acts on } \sin(\theta) \\ \text{term}}} + \underbrace{\frac{1}{\sin^2(\theta)} \frac{\partial^2 \psi}{\partial \phi^2}}_{\substack{\text{only acts on} \\ e^{-i\phi} \text{ term}}} \right]$$

term by term:

$$(1) -\hbar^2 C r e^{-kr} e^{-i\phi} \left[\frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial (\sin(\theta))}{\partial \theta} \right) \right]$$

$$\frac{\partial \sin(\theta)}{\partial \theta} = \cos(\theta)$$

$$= -\hbar^2 C r e^{-kr} e^{-i\phi} \left[\frac{1}{\sin(\theta)} \left(\frac{\partial}{\partial \theta} (\sin(\theta) \cos(\theta)) \right) \right]$$

$$\downarrow$$

$$\frac{\partial}{\partial \theta} (\sin(\theta) \cos(\theta)) = \frac{\partial}{\partial \theta} (\sin(\theta)) \cos(\theta) + \sin(\theta) \frac{\partial}{\partial \theta} (\cos(\theta))$$

$$= (\cos^2(\theta) - \sin^2(\theta))$$

• Can simplify $\cos^2(\theta) - \sin^2(\theta)$

$$\cos^2(\theta) + \sin^2(\theta) = 1 \Rightarrow \cos^2(\theta) = 1 - \sin^2(\theta)$$

thus,

$$\cos^2(\theta) - \sin^2(\theta) = (1 - \sin^2(\theta)) - \sin^2(\theta)$$

$$= (1 - 2\sin^2(\theta))$$

Plugging this back in

$$(1) = -\hbar^2 C r e^{-kr} e^{-i\phi} \left[\frac{1}{\sin(\theta)} (1 - 2\sin^2(\theta)) \right]$$

$$= -\hbar^2 C r e^{-kr} e^{-i\phi} \left(\frac{1}{\sin(\theta)} \right) + 2\hbar^2 \underbrace{C r e^{-kr} e^{-i\phi} \sin(\theta)}_{\psi}$$

$$(1) = -\hbar^2 C r e^{-kr} e^{-i\phi} \frac{1}{\sin(\theta)} + 2\hbar^2 \psi$$

Now, the second term

$$(2) = \frac{-\hbar^2}{\sin^2(\theta)} \frac{\partial^2 \psi}{\partial \phi^2} = \frac{-\hbar^2}{\sin^2(\theta)} (C r e^{-kr} \sin(\theta)) \frac{\partial^2 (e^{-i\phi})}{\partial \phi^2}$$

$$= \left(\frac{-\hbar^2}{\sin(\theta)} \right) (C r e^{-kr} (-i)^2 e^{-i\phi})$$

$$(2) = \frac{\hbar^2}{\sin(\theta)} (C r e^{-kr} e^{-i\phi})$$

these terms cancel each other

$$\text{Thus: } \vec{L}^2 \psi = (1) + (2) = 2\hbar^2 \psi$$

$$\vec{L}^2 \psi = 2\hbar^2 \psi$$

$$\hookrightarrow l(l+1)\hbar^2 = 2\hbar^2$$

so

$$l = 1$$

So, we found:

$$\hat{L}_z \psi = -\hbar \psi$$

$$\vec{L}^2 \psi = 2\hbar^2 \psi$$

$$\hat{L}_z \psi = m\hbar \psi$$

$$\vec{L}^2 \psi = \hbar^2 l(l+1) \psi$$

and

$$m = -1$$

$$l = 1$$

⑤ Now, $\hat{H}\psi = E_n \psi = -\frac{(\text{stuff})}{n^2} \psi$ $E_n = -\frac{13.6 \text{ eV}}{n^2}$

$\hat{H}$ is given on pg 3:

$$\hat{H} = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\vec{L}^2}{2\mu r^2} - \frac{\hbar^2}{a_0 \mu} \frac{1}{r}$$

Thus,

$$\hat{H}\psi = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{\vec{L}^2 \psi}{2\mu r^2} - \frac{\hbar^2}{a_0 \mu} \frac{\psi}{r}$$

$\downarrow$ ① $\downarrow$ we already found $\vec{L}^2 \psi$ above: $\downarrow$ Nothing to do here
 This term will act on $(r e^{-kr})$

$$\vec{L}^2 \psi = 2\hbar^2 \psi$$

* Look at term ①: $= -\frac{\hbar^2}{2\mu r^2} (\sin(\theta) e^{-i\phi} \frac{\partial}{\partial r} (r^2 \frac{\partial}{\partial r} (r e^{-kr})))$

$$\left(\frac{\partial}{\partial r} (r e^{-kr}) \right) = \frac{\partial r}{\partial r} e^{-kr} + r \frac{\partial e^{-kr}}{\partial r} = e^{-kr} + r(-k) e^{-kr}$$

$$= (1 - rk) e^{-kr}$$

Plug this in: $-\frac{\hbar^2}{2\mu r^2} (\sin(\theta) e^{-i\phi} \frac{\partial}{\partial r} (r^2 (1 - rk) e^{-kr}))$

$$\frac{\partial}{\partial r} (r^2 - r^3 k) e^{-kr}$$

$$\begin{aligned} \frac{\partial}{\partial r} (r^2 - r^3 k) e^{-kr} &= \left(\frac{\partial}{\partial r} (r^2 - r^3 k) \right) e^{-kr} + (r^2 - r^3 k) \frac{\partial e^{-kr}}{\partial r} \\ &= (2r - 3r^2 k) e^{-kr} + (r^2 - r^3 k) (-k) e^{-kr} \\ &= e^{-kr} [2r - 3r^2 k - kr^2 + r^3 k^2] \\ &= e^{-kr} [r^3 k^2 - 4r^2 k + 2r] \end{aligned}$$

$$(1) = -\frac{\hbar^2}{2\mu r^2} (\sin(\theta) e^{-i\phi} [r^3 k^2 - 4r^2 k + 2r] e^{-kr})$$

factor out an r

$$= -\frac{\hbar^2}{2\mu r^2} (r \sin(\theta) e^{-i\phi} e^{-kr}) [r^2 k^2 - 4rk + 2]$$

$$= \frac{-\hbar^2}{2\mu} \psi \left[\frac{r^2 k^2}{r^2} - \frac{4rk}{r^2} + \frac{2}{r^2} \right]$$

$$(1) = \frac{-\hbar^2}{2\mu} (4) \left[k^2 - \frac{4k}{r} + \frac{2}{r^2} \right]$$

Now, Remember we want $H\psi = E_n \psi$, so the final terms with $\frac{1}{r}$ and $\frac{1}{r^2}$ should cancel the other two terms in the Hamiltonian above

$$\hat{H}\psi = (1) + \frac{L^2\psi}{2\mu r^2} - \frac{\hbar^2\psi}{a_0\mu r}$$

$$= \left[\frac{-\hbar^2\psi k^2}{2\mu} + \frac{\hbar^2 4k\psi}{2\mu r} - \frac{\hbar^2 2\psi}{2\mu r^2} \right] + \frac{L^2\psi}{2\mu r^2} - \frac{\hbar^2\psi}{a_0\mu r}$$

Recall $L^2\psi = 2\hbar^2\psi$, so

$$\hat{H}\psi = \left[\frac{-\hbar^2\psi k^2}{2\mu} + \frac{\hbar^2 4k\psi}{2\mu r} - \frac{\hbar^2 2\psi}{2\mu r^2} \right] + \frac{2\hbar^2\psi}{2\mu r^2} - \frac{\hbar^2\psi}{a_0\mu r}$$

cancel

These must cancel

$$\Rightarrow \frac{\hbar^2 4k}{2\mu r} = \frac{\hbar^2}{a_0\mu r}$$

$$\Rightarrow 2k = \frac{1}{a_0} \Rightarrow$$

$$k = \frac{1}{2a_0}$$

* In general, $k = \frac{1}{na_0}$, so that means

$$n = 2$$

This leaves us with

$$\hat{H}\psi = -\frac{\hbar^2 k^2}{2\mu} \psi = -\frac{\hbar^2}{2\mu} \left(\frac{1}{2a_0} \right)^2 \psi = -\left(\frac{\hbar^2}{2a_0^2\mu} \right) \frac{1}{2^2} \psi$$

$$\hat{H}\psi = -\left(\frac{\hbar^2}{2a_0^2\mu} \right) \frac{1}{2^2} \psi$$

Note we found our three quantum numbers

$$m = -1$$

$$l = 1$$

$$\Rightarrow |2, 1, -1\rangle \text{ or } \psi_{2,1,-1}$$

$$n = 2$$

(Can check pg 99 of notes to see $u_{2,1,-2}$)

③ find from

$$1 = \int |\psi|^2 dV = \int_0^\infty \int_0^{2\pi} \int_0^\pi \underbrace{\psi^* \psi r^2 \sin(\theta)}_{\text{These are the bounds for each variable so,}} d\theta d\phi dr$$

These are the bounds for each variable so,

$$0 \leq r < \infty$$

$$0 \leq \phi \leq 2\pi$$

$$0 \leq \theta \leq \pi$$

(see pg 99 to check your answer for your constant C .)

④ Recall: $\langle r \rangle = \langle \psi | \hat{r} | \psi \rangle = \int |\psi|^2 r dV$

or

$$\langle r \rangle = \int_0^\infty \int_0^{2\pi} \int_0^\pi (\psi^* \psi r) r^2 \sin(\theta) d\theta d\phi dr$$

and likewise

$$\langle r^2 \rangle = \int_0^\infty \int_0^{2\pi} \int_0^\pi (\psi^* \psi r^2) r^2 \sin(\theta) d\theta d\phi dr$$

Δr is given by

$$\Delta r = \sqrt{\langle r^2 \rangle - \langle r \rangle^2}$$

⑤ Think about energy differences $\Delta E = |E_f - E_i|$

